

Caleb James Smith

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Employment

AI Evaluator (Part Time)	Data Annotation	Feb. 2026–Present
Postdoctoral Researcher	The University of Kansas	Jan. 2021–Jan. 2026
Research Assistant	Baylor University	May 2016–Dec. 2020
Teaching Assistant	Baylor University	Aug. 2014–May 2016
Undergraduate Researcher	Baylor University	May 2013–Aug. 2013

Education

PhD in Physics	Baylor University	Aug. 2014–Dec. 2020
MA in Physics	Baylor University	Aug. 2014–May 2018
BS in Physics, Minor in Mathematics	Taylor University	Aug. 2011–May 2014

Computing Skills

Operating Systems	macOS, Linux, Windows
Languages	Python, C++, Bash, HTML, CSS
Libraries	Matplotlib, NumPy, Pandas, Flask, Django, ROOT
Tools	Unix/Linux Terminal, VS Code, Vim, Git, JSON, XML, Markdown, $\LaTeX$, Claude Code, Jupyter Notebook, Mathematica, Microsoft Office

Awards

Breakthrough Prize in Fundamental Physics, 2025! 🏆

Texas Section of the American Physical Society (TSAPS) Graduate Student Presentation Award, 2020.

Outstanding Physics & Engineering Senior, Taylor University, 2014.

Louis Armstrong Award, Taylor University Jazz Ensemble, 2013 and 2014.

Advanced Placement (AP) Scholar with Honor Award, College Board, 2010.

Fellowships

DOE Office of Science Graduate Student Research (SCGSR) Program at Fermilab – 2018 to 2019.

LHC Physics Center (LPC) Guests and Visitors Program at Fermilab – 2016, 2018 and 2019.

Baylor University Graduate School Fellowship – 2014 to 2019.

Research

Postdoctoral

CMS tracker Phase 2 upgrade (KU, 2021–Present)

The CMS inner tracker will be replaced in the Phase 2 upgrade. Over two billion pixels will be installed, and custom RD53 chips will process and encode pixel data. High bandwidth electrical and optical links will be used to provide communication between the RD53 chips and off detector electronics. At the University of Kansas (KU), I am implementing tests for the high speed electrical links in this readout chain. We use a variety of setups to test electrical links for continuity, measure bit error rates, produce eye diagrams, measure crosstalk between channels, and measure impedances. These measurements provide robust tests of electrical link quality.

CMS compressed SUSY analysis (KU, 2021–Present)

Supersymmetry (SUSY) is an extension of the standard model (SM) that can explain many open questions in particle physics, including the hierarchy problem and the fundamental description of dark matter. The CMS experiment continues to explore new regions of phase space to search for evidence of supersymmetric particles. Our group at KU is focusing on compressed SUSY models that have low momentum objects (leptons, quarks, etc.) in the final state. I performed studies regarding low momentum bottom quarks and electrons. Improving low momentum object identification increases the search sensitivity for various SUSY models.

CMS pixel operations (CERN, 2023)

I worked with the CMS pixel operations team at CERN. I took pixel detector on-call (DOC) shifts, which included monitoring detector power, temperatures, and other parameters, performing calibration runs, and responding quickly to solve detector problems. In addition, I helped to develop a new front-end driver (FED) monitoring website. The CMS pixel detector has 108 FEDs that are used to collect data from readout chips on the pixel detector. The new FED monitoring website provides live monitoring data for all pixel FEDs, which helps quickly pinpoint FED problems during CMS data-taking.

CMS tracker DPG (KU, 2021–2022)

Tracking and alignment of the CMS detector depend on measuring a precise beam spot of the proton-proton collisions provided by the LHC near the center of CMS. Typically, the beam spot is measured using tracking information and reconstructed tracks. However, track reconstruction requires significant computing time. I worked on a novel fitting method to predict the beam spot location without using reconstructed particle tracks. Various methods were developed and tested on simulation and data.

Graduate

CMS SUSY analysis (Fermilab, 2018–2020)

There are many CMS analysis groups searching for different forms of new physics. I worked on a search for supersymmetry (SUSY) using the Run 2 dataset from CMS. We looked for top squarks that decay into all-hadronic (zero lepton) final states. We specifically targeted nine simplified SUSY models including T2tt and T1tttt. I focused on predicting the Z invisible background that accounts for events in which a Z decays to two neutrinos and results in missing transverse energy.

CMS HCAL Phase 1 upgrade installation and commissioning (CERN, 2016–2018)

In the fall of 2016, I went to CERN where I supported testing and commissioning of Hadron Calorimeter (HCAL) front-end electronics for the HCAL Endcap (HE) Phase 1 upgrade. Under the supervision of Jim Hirschauer from FNAL, I managed quality assurance tests for HE calibration units (CU) at CERN. The calibration units are used to send either LED or laser light to the primary light detectors, silicon photomultipliers (SiPM), and can be used to tune SiPM gains. I also participated in initial commissioning of the HCAL Barrel (HB) readout electronics at CERN, including testing new HB backplanes and new front-end communication software, measuring muon data with HB Readout Modules (RM) in testbeam, and performing HB RBX power and cooling tests.

CMS HCAL online software (CERN, 2017–2018)

I joined the HCAL Online Software (HCOS) group in 2017. This required being an on-call expert for CMS HCAL online software issues in week-long shifts. I helped solve software problems in the production CMS setting to ensure quality data-taking. Additionally, I developed software for the configuration and monitoring of the Phase 1 upgrade readout electronics.

CMS HCAL Phase 1 upgrade QIE card testing (Fermilab, 2016 and 2018)

The CMS HCAL Phase 1 upgrade involved installing silicon photomultipliers (SiPM) to measure light from scintillators in the detector. The SiPM signals are digitized by custom readout cards, called QIE cards, using the FNAL-designed Charge Integrator and Encoder version 11 (QIE11) chip. In 2016, I created software to test the hardware and firmware functionality of about 750 HE QIE cards and contributed to the development of a database to save the test results and calibration constants for all cards. In 2018, I helped with testing about 880 HB QIE cards by implementing an efficient method to automatically upload firmware to multiple QIE cards in series.

Undergraduate

Data encryption using coupled chaotic circuits with Dr. Ken Kiers (Taylor University, January 2014).

String theory model building with Dr. Gerald Cleaver (Baylor University, CASPER, Summer 2013).

Teaching**Baylor Physics 1409 Lecture (Electricity and Magnetism, algebra based), 2015–2016**

I taught an introductory electricity and magnetism physics course. I prepared lecture material, taught the lectures, assigned online homework problems, and prepared and graded exams. I enjoyed giving lectures, explaining physics concepts, and working through example problems with the class.

Baylor Physics 1409/1430 Lab (Electricity and Magnetism), Summer 2015

I taught introductory electricity and magnetism physics labs. This included explaining the physics theory and the lab procedure, helping students perform the experiment, and grading the lab reports. I enjoyed helping students understand and demonstrate physics concepts in the lab experiments.

Baylor Physics 1408/1420 Lab (Classical Mechanics), 2014–2015

I taught introductory classical mechanics physics labs. This included explaining the physics theory and the lab procedure, helping students perform the experiment, and grading the lab reports. I enjoyed helping students understand and demonstrate physics concepts in the lab experiments.

Selected Publications

Coauthor on over 800 papers published by the CMS Collaboration; list available on Inspire HEP:
<https://inspirehep.net/authors/1398495>

General search for supersymmetric particles in scenarios with compressed mass spectra using proton-proton collisions at $\sqrt{s} = 13$ TeV

V. Chekhovsky *et al.* (CMS Collaboration), Phys. Rev. D 112, 112023, December 29, 2025
<https://doi.org/10.1103/b26z-zmpy>

Characterization of a high bandwidth readout chain for the CMS Phase-2 pixel upgrade

C. Smith on behalf of the CMS tracker collaboration, 2022 JINST 17 C08017, August 23, 2022
<https://doi.org/10.1088/1748-0221/17/08/C08017>

Search for top squark production in fully hadronic final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

A. M. Sirunyan *et al.* (CMS Collaboration), Phys. Rev. D 104, 052001, September 10, 2021
<https://doi.org/10.1103/PhysRevD.104.052001>

Search for supersymmetric top quarks in the CMS Run 2 data set

C. Smith, PhD Dissertation, Baylor University, October 21, 2020
<https://baylor-ir.tdl.org/handle/2104/11191>

The Phase-2 upgrade of the CMS barrel calorimeters

CMS Collaboration, CMS Technical Design Report, September 12, 2017
<https://cds.cern.ch/record/2283187>

Selected Presentations

Development of a high bandwidth readout chain for the CMS Phase-2 pixel upgrade

TWEPP 2021¹, September 21, 2021

<https://indico.cern.ch/event/1019078/contributions/4444260>

Searches for third generation squarks with the CMS detector

SUSY 2021¹, August 23, 2021

<https://indico.cern.ch/event/875077/contributions/4485659>

Search for Supersymmetric Top Quarks in the CMS Run 2 Data Set

Texas Section APS 2020¹, November 13, 2020

<https://tsapsf20.uta.edu/agenda-nov-13>

Search for Supersymmetric Top Quarks in the CMS Run 2 Data Set

PhD Dissertation Oral Presentation¹, October 9, 2020

<https://indico.cern.ch/event/956156/#1-search-for-supersymmetric-to>

Search for Supersymmetric Top Squark Production at CMS with Heavy Object Taggers

APS April 2020¹, April 18, 2020

<http://meetings.aps.org/Meeting/APR20/Session/B13.5>

Phase 1 Upgrade for the CMS Hadron Endcap Calorimeter

2018 US LHC Users Association Meeting at Fermilab, October 26, 2018

<https://indico.fnal.gov/event/17566/session/2/contribution/75/material/slides/0.pdf>

HCAL Endcap Installation and Commissioning

LPC Physics Forum at Fermilab, August 30, 2018

<https://indico.cern.ch/event/744809/#1-hcal-endcap-installation-and>

Completing the CMS HCAL Endcap Phase 1 Upgrade (Poster)

51st Annual Users Meeting at Fermilab, June 20, 2018

<https://indico.fnal.gov/event/16332/session/10/contribution/114/material/poster/0.pdf>

Online Software and Monitoring

HCAL meetings during CMS week: PM report and Operations session, April 16, 2018

<https://indico.cern.ch/event/719790/#5-online-software-and-monitori>

Phase 1 Frontend Monitoring, Plans for 2018

HCAL meetings during CMS week: PM report, HB Phase 1 summary, and Operations session
February 5, 2018

<https://indico.cern.ch/event/701182/#4-phase-1-frontend-monitoring>

HE Calibration Unit Validation

HCAL meetings during CMS week: HB and HE Phase1 session, December 3, 2017

<https://indico.cern.ch/event/683888/#1-he-calibration-unit-validati>

HE Calibration Unit Characterization

HCAL meetings during CMS week: HB and HE Phase1 session, April 2, 2017

<https://indico.cern.ch/event/624907/#7-he-calibration-unit-characte>

¹Virtual meeting due to COVID-19.

References

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